

# Assignment 4

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## Introduction

For this assignment, I will be attempting to build a tiny language model using an LSTM trained on some text from Project Gutenberg, in this case *Romeo and Juliet*, as I'm curious to see if a poorly trained, tiny, outdated-architecture language model can imitate Shakespeare. Later, I will experiment with a pretrained Transformer model, in this case GPT-2, and examine the differences between the two.

## Analysis

To start with the first part of the assignment, the first step was to gather and clean my data. Thus, I browsed Project Gutenberg's most frequently downloaded texts, and was *going* to use *Frankenstein*, but ultimately settled on *Romeo and Juliet* as I was intrigued by how the LSTM would react to this tone. I then defined when the text scan would start and end (had to make sure to cut out the credits and table of contents, and end before the metadata), and split it into words, of which there were approximately 26000. Among these 26000 words, there were around 5726 unique words, which is interesting, because in the in class example, there were nearly 5 times as many words, but only around twice as many unique words when compared with *Romeo and Juliet*. I then generated embeddings for each unique word and split them into sequences. Then, I defined the sequence length, shortening it from 20 to 15 due to having far less text available than the in-class example.

For the second part of the assignment, all I really did was load in GPT-2 and offer it the prompt, "In my opinion, machine learning is", due to having seen it offer some radical opinions when offered other prompts about people, places, and things. I kept the temperature at 1 but increased the max token output to 50.

## Methodology

For the first part, I defined the LSTM network, with a few key changes. As previously mentioned, I'm working with less data, so I decreased the sequence length, thereby increasing the number of sequences fed into the network. However, I still didn't want to have the network overfit at all, so I increased the dropout to 0.4. I was going to adjust the number of recurrent units or embedding dimensions (probably decreasing them since the dataset was technically much more simple), but considered that since powers of 2 were customarily used for these parameters, I didn't want to fully cut them in half to 256 and 128, respectively, as that felt too drastic of a change, so ultimately I kept them the same.

For the Transformer, GPT-2 is a Decoder-only Transformer model developed by OpenAI, which has multiple variants that determine the parameter size, but the one I used, the "standard" version, has 124 million parameters (and is the smallest of the 3). This is a stark contrast to my

LSTM model, which in addition to being an inferior architecture for any language modeling task, also only had around 22 million parameters and much less training.

## Results

For the LSTM model, the final text generation was as follows, given the starting prompt “romeo felt”, 500 words to predict, at a temperature of 1:

“romeo felt for envious? partizans.\_] come descry. money, simple hair  
chambermaids. wise, night; merchandise. hearts, heel wake, benvolio. for settled  
gentler kiss wounded rocks church, nose passado, fright notice plate. [\_ enters the  
affords you, underneath a cave? pump, thou strew take a anger bring likewise  
cats. thin whate’er married doth on my maidenheads; ready? her am: thy hand  
wife. festival affections’ ensign montague, the word. potpan! sweet. faith, come  
case. fetch early? traces, part!’ i i pale lineament, partisans, leg cannot. nine beat  
delay to arm heap’d it then this riband? madness sighs, text. window, i am green  
youth again; can help! we black mend. action here or our hoodwink’d crutch!  
points, drawn, it,—here should hear put pathway, what hast prevails then. proof  
alla learned. in out was thou, make body’s fly. exhales gentle page? perchance  
mercutio’s which that it will one unattainted her live paradise, younger darkness  
with have devise like, afraid paris your she,— heaven, nightingale, lure they she  
is is a valentine; that i have unattainted beauty, is the wedding knaves; know;  
join-stools, madman! off, physic sweeter present tops,— scratch crossing lace  
then were himself a stain’d time arm than earth? poison, in are away: ghost trunk  
thou that likely fetch romeo? greatest, will the present bride moveth subject say’t  
troth, the fish francis! babe done. play, faith? did poor flattering word’s i were  
bad? must brine revolts lucentio, friends; hands? comes. sluttish leap scene.  
ignorance, disturb’d lawrence.\_] calm dead, nature, affection tasted, rite, learns  
are: eye wise dinner noble haunt tomorrow sad tybalt? sight. mood waking, let my  
secret? sought prating [\_kisses stay, hands fright ever, loggerhead.—good wit,  
signify bad speaks breast. weeping head romeo. romeo. for miss, cries,  
unaccustom’d close, if slain, take that he shall black, ha, her, romeo. wife. sound.  
three worm fortune! romeo! ended passion! overlooking men, death with her  
comfort; hated, romeo’s, quiet, bawd, these vile tonight? burial heaven. heaven.  
business, thirty toad that paris bliss o say. ill-beseeming juliet.\_] ‘i’ll serve for or  
he new kill’d. cheeks, long we’ll me the eternal years well making disparagement.  
pilcher sin [\_kissing cast valour door amen. matter. lead i say! care of a without  
shall i will matter fairly monument.\_] simple cannot advanc’d, soft! and at us, i  
paris. you, present needs prove began? unworthiest will. myself, iii lady we,  
broke, have all? like it whose rest what have what men’s silver myself. abate me  
nurse. sin. seen, brief, capulet, of thee back blood, fay, heard man. hadst faith  
cancell’d asleep! lies, skillless an do wine. what’s neither, place, being thought

place. let's church; herbs, wit, religion capulet! as much, liege, thy drops back  
stranger revolt eye: sweet, did shall wilt stay back dry-beat with kill'd, depart  
stand. lend purpose. mother?' unluckily benvolio. and nurse. with let bosom looks  
of her lie beauty death; greater own some forfeit everyone spite vexed upon scape  
she, anything have to with blows leg, wish. soul, piercing it.\_] presses cat, first  
lest tybalt lord, to sun star-cross'd peter.\_] reputation"

Now, while there is a possibility that my mortal mind is simply incapable of understanding some hidden brilliance or meaning in this passage, it's far more likely that this model is just delusionally hallucinating due to a lack of training data variety, bad embedding strategy (no sense of context awareness), and poor overall training. The results from GPT-2 were far more interesting:

"In my opinion, machine learning is a very good way to learn. It's not a perfect way to learn, but it's a good way to learn.

I think machine learning is a very good way to learn. It's not a perfect way to learn, but it"

It might just be me, but my reaction to this generation was twofold. Firstly, I was shocked at how mellow its response was, as it had shown no mercy in absolutely tearing into prominent political figures, celebrities, and other people in the past. My second reaction, however, was that the tone of this (with all the repetition and need to emphasize the superior nature of the subject of the statement) reminded me very much of the manner of speech of our current president. I suspect this is because it essentially finished all it needed to say by the end of the first sentence, and was just generating tokens to meet the 50-token parameter I set earlier.

## Reflection

Throughout this assignment, I learned a few valuable lessons. One was the importance of data, which really put into perspective the absurd amounts of money companies are spending on data these days, and why companies like Google have such a head start in the AI race. I look forward to experimenting further with LSTMs, since as we've seen (through models such as HRM), they aren't exactly irrelevant.